Practice workbook · What encryption is and why we use it

What encryption is and why data is encrypted.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-17

1. Question 3 · part b · 1 mark

2020 NOV · Paper 02

Worked-example retry · Attempt part b only.

3 Several encryption algorithms have been developed.

(a) State what is meant by the term 'encryption'.

(1)

(b) Give **one** reason why data may need to be encrypted.

(1)

(c) (i) Perform a Rail Fence cipher encryption of the text **COMPUTATIONAL** using a key of 4.

Show your working.

(4)

Encrypted text:

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 1 8 8 5 A 0 6 1 6

1. Question 3 · part b · 1 mark · Working and answer

Working and answer

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2. Question 6 · part a.iii · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part a.iii only.

6 A train company uses ticket vending machines at each station.

(a) The machines use embedded systems.

(i) Explain **one** benefit of using an embedded system in these machines.

(2)

(ii) Customers use a touch screen to select their destination. They can pay by cash or bank card. Their tickets and a receipt are printed.

The touch screen is controlled by an embedded system.

Give **two** other hardware components in the ticket machine that are controlled by embedded systems.

(2)

1 .....

2 .....

(iii) The ticket machine uses data encryption when a customer pays using a bank card.

State why data encryption is used in this case.

(1)

.....

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

20

P 6 1 8 8 0 A 0 2 0 2 4

2. Question 6 · part a.iii · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part a.iii only.

(b) Compare **four** features of high-level and low-level programming languages.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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2. Question 6 · part a.iii · 1 mark · Working and answer

Working and answer

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3. Question 3 · part a · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part a only.

3 Several encryption algorithms have been developed.

(a) State what is meant by the term 'encryption'.

(1)

(b) Give **one** reason why data may need to be encrypted.

(1)

(c) (i) Perform a Rail Fence cipher encryption of the text **COMPUTATIONAL** using a key of 4.

Show your working.

(4)

Encrypted text:

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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3. Question 3 · part a · 1 mark · Working and answer

Working and answer

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4. Question 4 · part a · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a only.

4 Several encryption algorithms have been developed.

(a) Identify what is meant by **encryption**.

(1)

- ☒ **A** Conversion of ciphertext into plaintext
- ☒ **B** Conversion of plaintext into ciphertext
- ☒ **C** Conversion of code into data
- ☒ **D** Conversion of data into information

(b) **Figure 1** shows a Pigpen cipher grid.

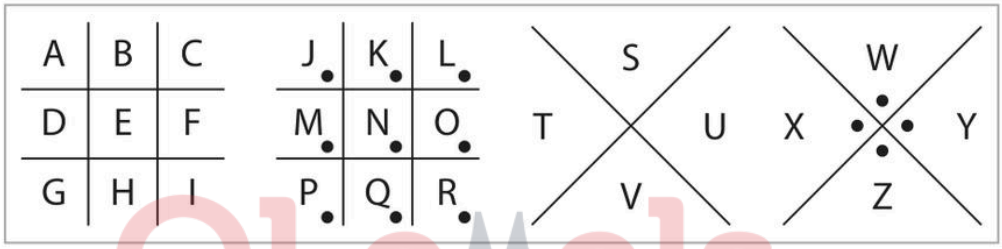


Figure 1

Complete the table by adding the symbol for each letter in the word **MAY**.

(3)

Letter	M	A	Y
Symbol			

(c) The message THE ENEMY IS NEAR is encoded using a different encryption algorithm.

Figure 2 shows the first stage of the encryption process.

T								I					
	H						Y		S				
		E				M				N			
			E		E						E		R
				N								A	

Figure 2

8

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4. Question 4 · part a · 1 mark · Working and answer

Working and answer

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5. Question 1 · part b.i · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.i only.

1 People use networks in all aspects of their lives.

(a) An office building has desktop computers connected in a network.

One reason for networking computers is to enable technicians to maintain them remotely.

State **two other** reasons for connecting computers in a network.

(2)

1 .....

.....

2 .....

.....

(b) Access to data must be controlled.

(i) Identify **one** reason that encryption is used.

(1)

☒ **A** To compress the file so it takes up less storage space

☒ **B** To make sure both sender and receiver use the same key

☒ **C** To make sure data is only understood by the intended receiver

☒ **D** To stop malicious hackers getting into a computer network

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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2

P 7 2 9 3 7 A 0 2 2 4

5. Question 1 · part b.i · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(ii) File servers are in a small room at the back of an office.

The office has a burglar alarm.

Give **two other** ways that the servers can be secured using physical methods.

(2)

1 .....

.....

2 .....

.....

(iii) Phishing is a type of social engineering.

State what is meant by the term **social engineering**.

(1)

.....

.....

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(c) 5G is a communication standard.

Identify the characteristic that is true for the 5G communication standard.

(1)

- ☒ **A** It has a high transmission latency
- ☒ **B** It has built-in security
- ☒ **C** It has a lower bandwidth than 3G and 4G
- ☒ **D** It uses wireless communication

5. Question 1 · part b.i · 1 mark · Working and answer

Working and answer

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